A: Datasheet

Algorithm: idemia_011

Developer: Idemia

Submission Date: 2024_03_05

Template size: 300 bytes

Template time (2.5 percentile): 822 msec

Template time (median): 834 msec

Template time (97.5 percentile): 852 msec

Investigation:

Frontal mugshot ranking 35 (out of 481) −− FNIR(1600000, 0, 1) = 0.0010 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 52 (out of 443) −− FNIR(1600000, 0, 1) = 0.0076 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 8 (out of 412) −− FNIR(1600000, 0, 1) = 0.0528 vs. lowest 0.0510 from nec_009

Immigration visa−border ranking 15 (out of 370) −− FNIR(1600000, 0, 1) = 0.0008 vs. lowest 0.0005 from psl_001

Immigration visa−kiosk ranking 5 (out of 314) −− FNIR(1600000, 0, 1) = 0.0402 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 6 (out of 481) −− FNIR(1600000, T, L+1) = 0.0011, FPIR=0.003000 vs. lowest 0.0010 from sensetime_009

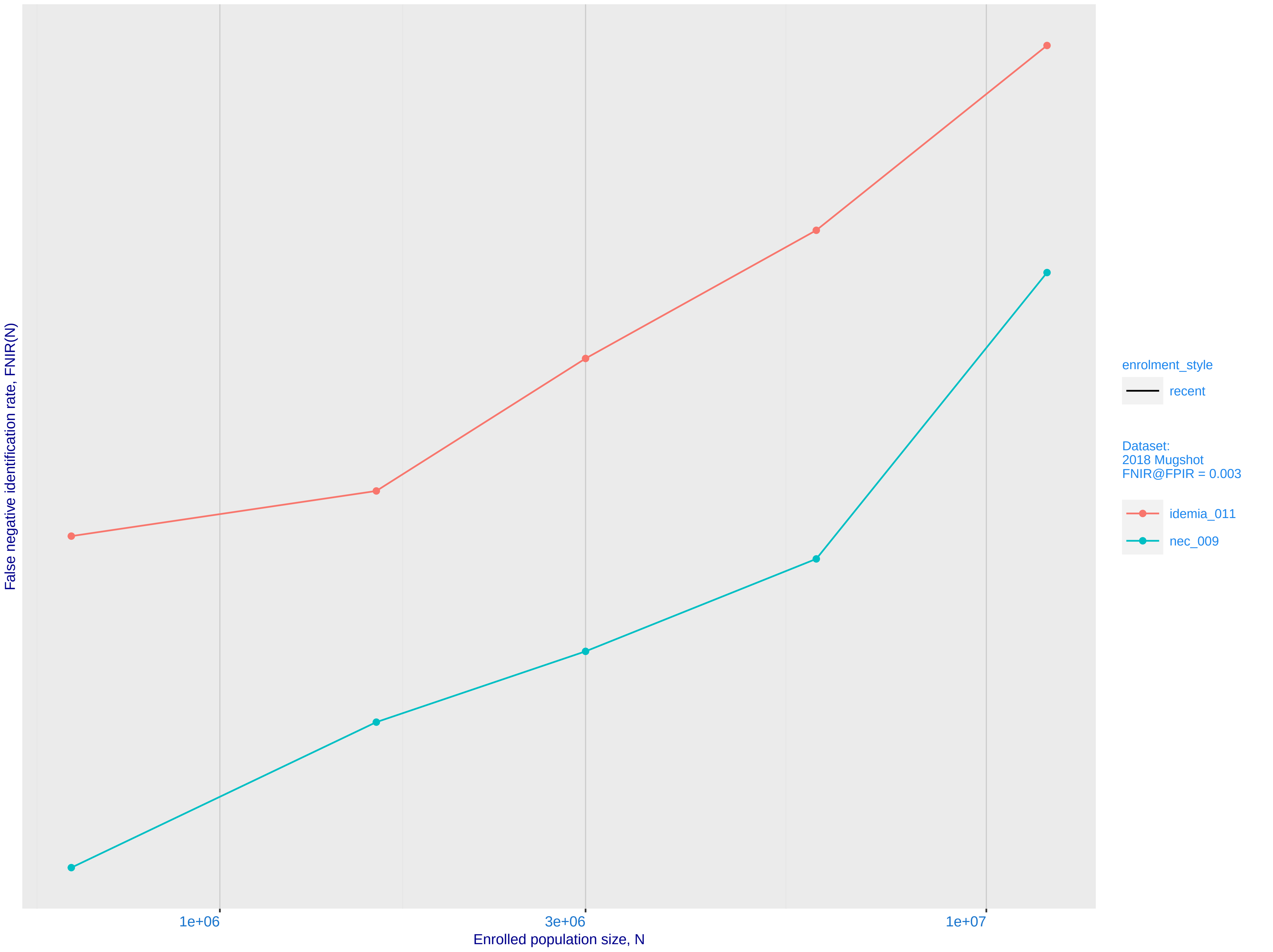
Mugshot webcam ranking 12 (out of 441) −− FNIR(1600000, T, L+1) = 0.0088, FPIR=0.003000 vs. lowest 0.0065 from sensetime_009

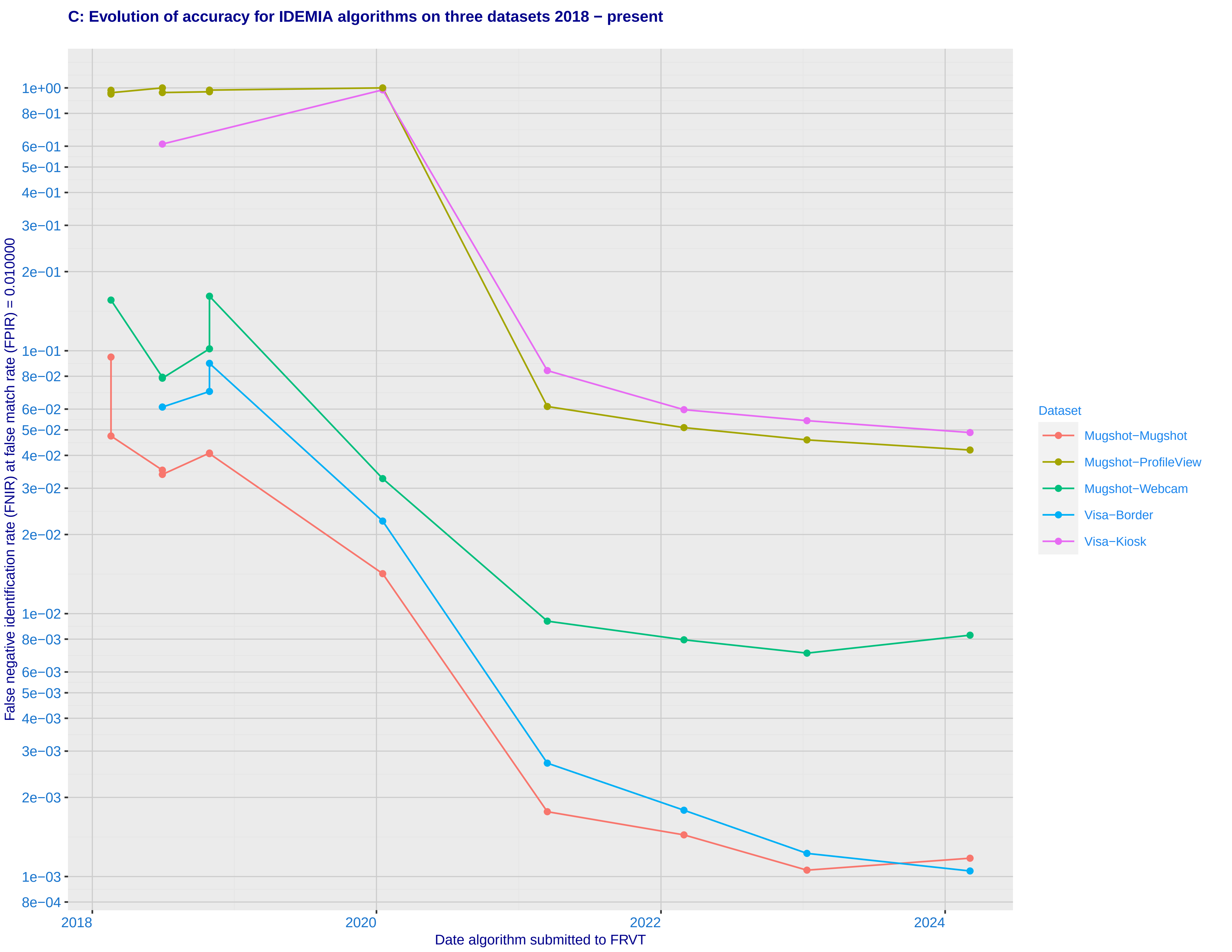
Mugshot profile ranking 4 (out of 411) −− FNIR(1600000, T, L+1) = 0.0672, FPIR=0.003000 vs. lowest 0.0591 from cloudwalk_mt_002

Immigration visa−border ranking 6 (out of 369) −− FNIR(1600000, T, L+1) = 0.0013, FPIR=0.003000 vs. lowest 0.0009 from cloudwalk_mt_002

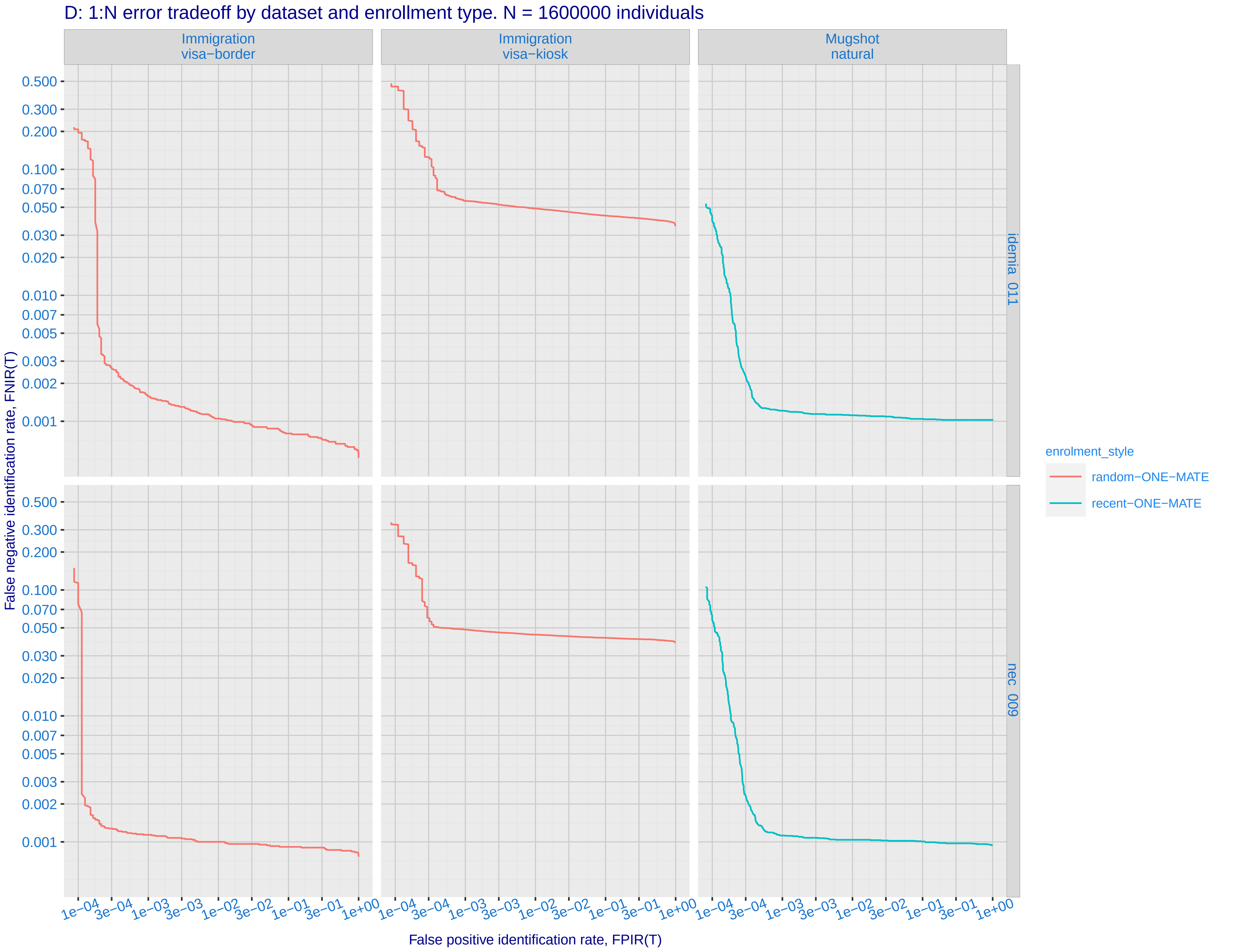
Immigration visa−kiosk ranking 5 (out of 314) −− FNIR(1600000, T, L+1) = 0.0524, FPIR=0.003000 vs. lowest 0.0460 from nec_009

B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (nec_009)

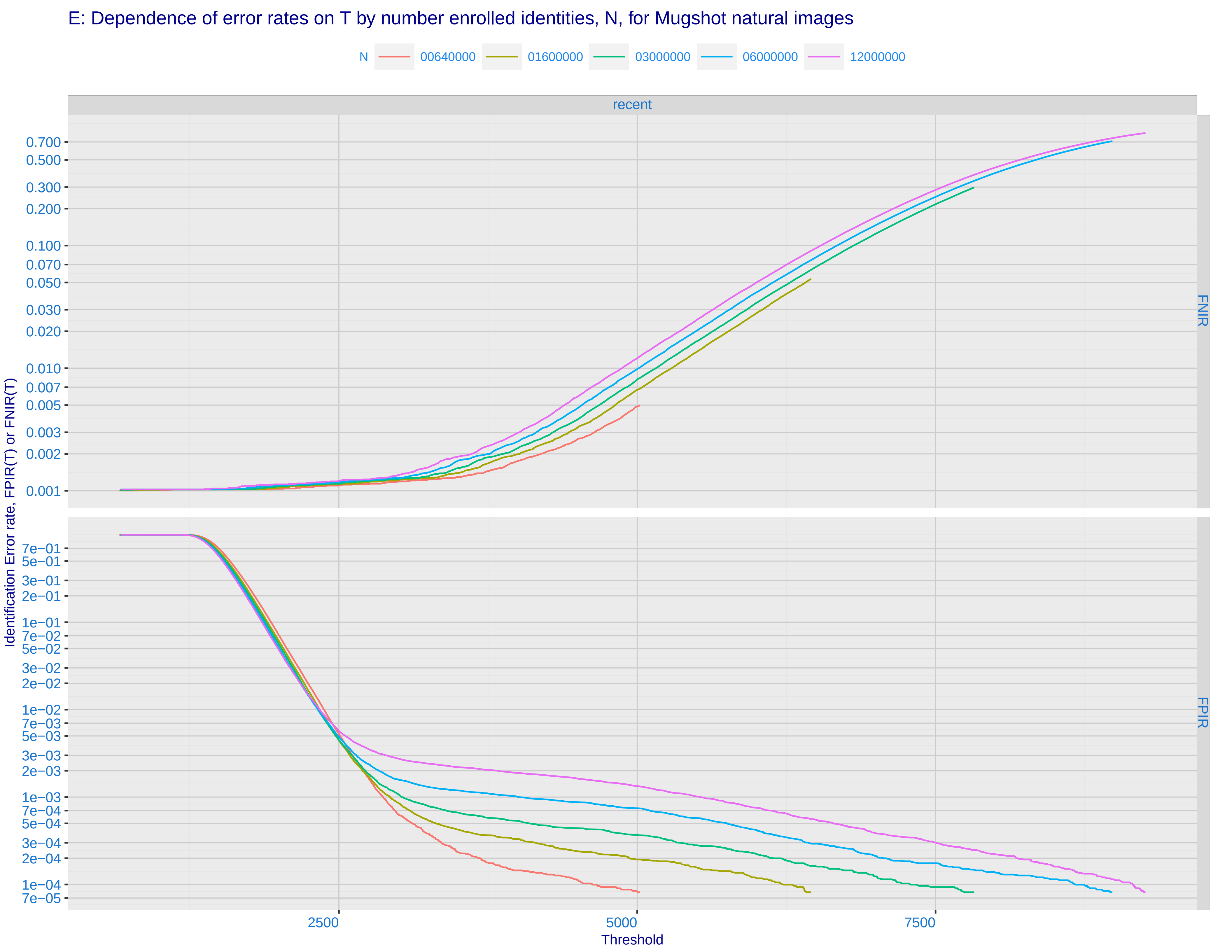


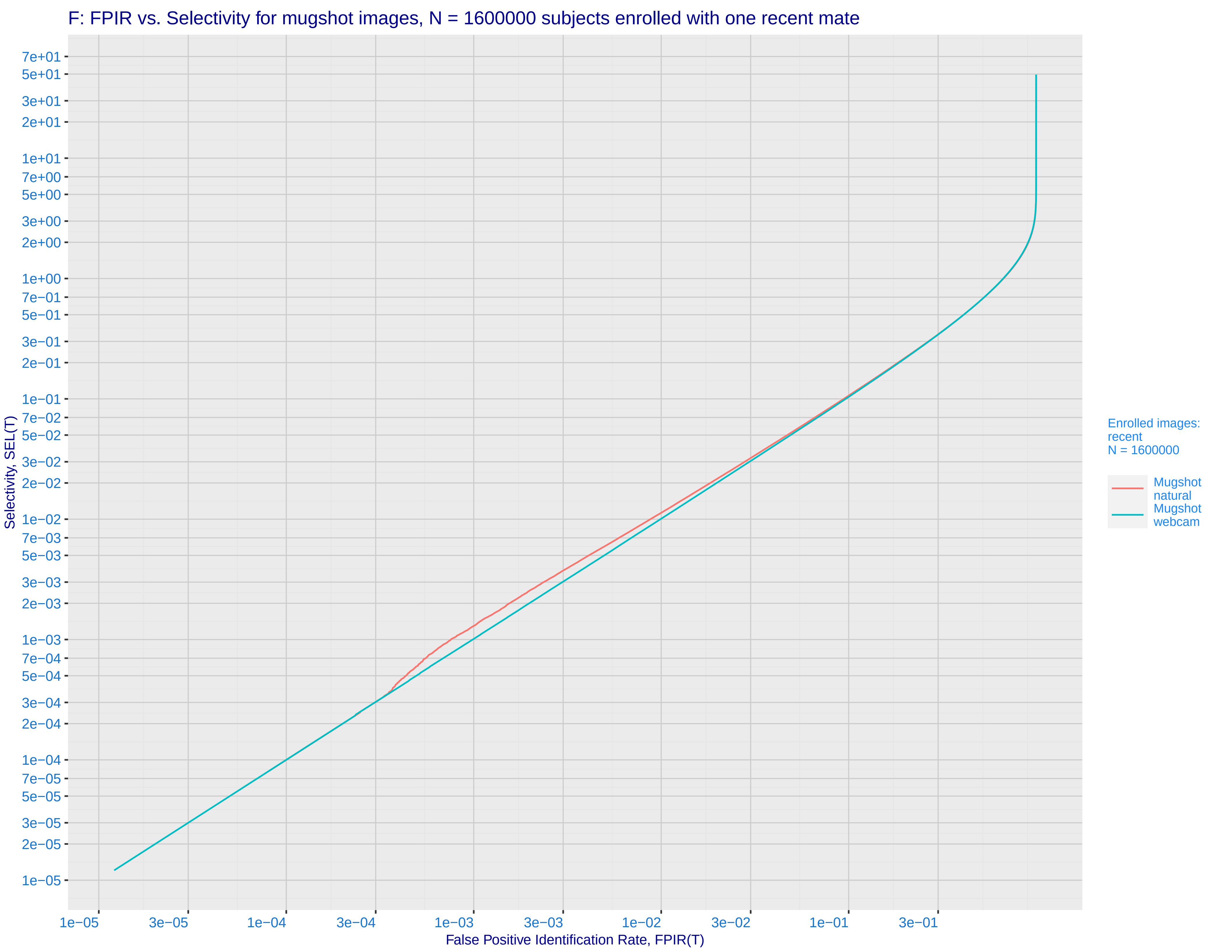


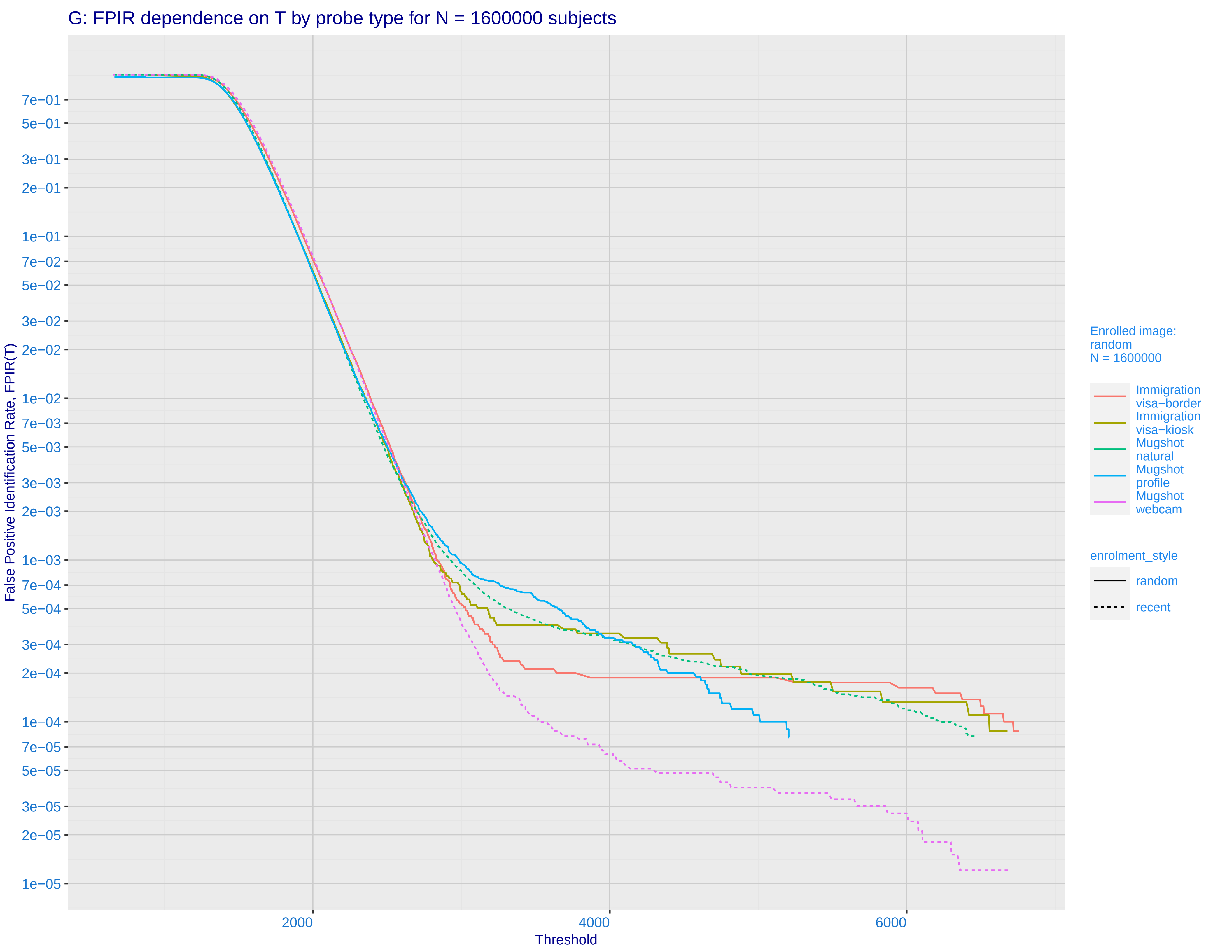
C: Evolution of accuracy for IDEMIA algorithms on three datasets 2018 − present
False negative identification rate (FNIR) at false match rate (FPIR) = 0.010000
1e+00
8e−01
6e−01
5e−01
4e−01
3e−01
2e−01
1e−01
8e−02
6e−02
5e−02
4e−02
3e−02
2e−02
1e−02
8e−03
6e−03
5e−03
4e−03
3e−03
2e−03
1e−03
8e−04
2018
2020
2022
2024
Date algorithm submitted to FRVT
Dataset
Mugshot−Mugshot
Mugshot−ProfileView
Mugshot−Webcam
Visa−Border
Visa−Kiosk

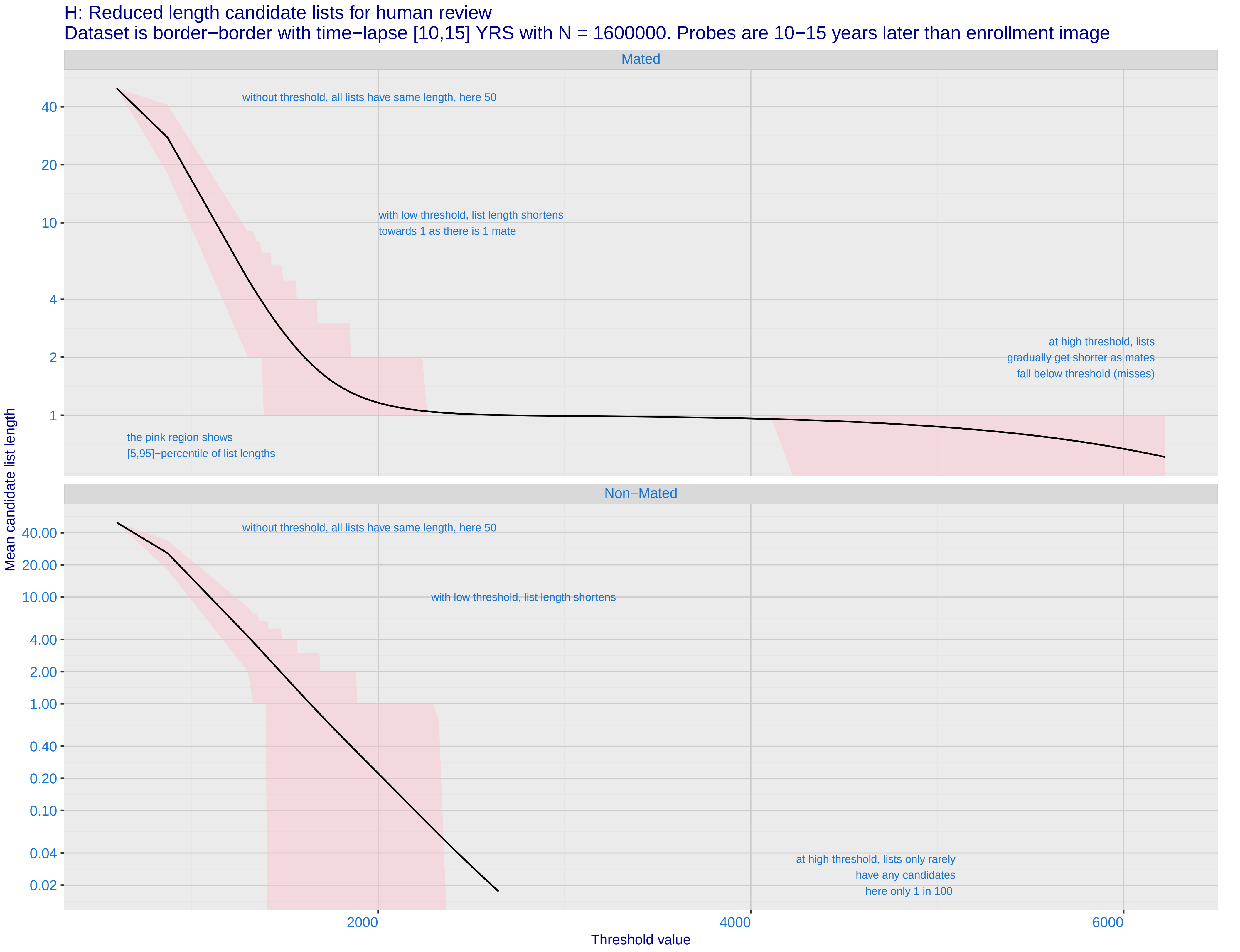


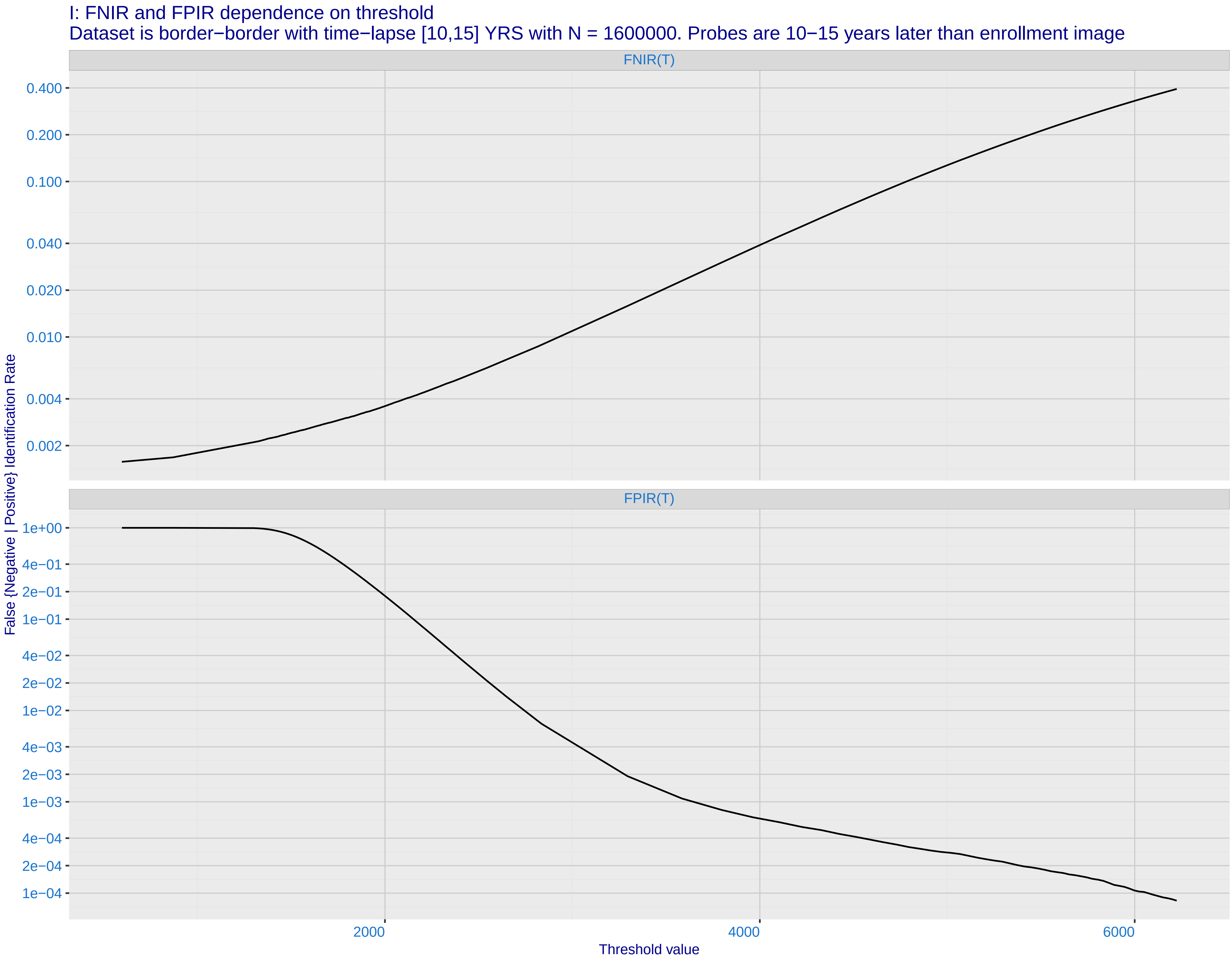
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals
Immigration
visa−border
Immigration
visa−kiosk
Mugshot
natural
idemia_011
nec_009
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(T)
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
1e+00
False positive identification rate, FPIR(T)
enrolment_style
random−ONE−MATE
recent−ONE−MATE


E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images
N
00640000
01600000
03000000
06000000
12000000
recent
FNIR
FPIR
Identification Error rate, FPIR(T) or FNIR(T)
Threshold
2500
5000
7500
0.001
0.002
0.003
0.005
0.007
0.010
0.020
0.030
0.050
0.070
0.100
0.200
0.300
0.500
0.700
7e−05
1e−04
2e−04
3e−04
5e−04
7e−04
1e−03
2e−03
3e−03
5e−03
7e−03
1e−02
2e−02
3e−02
5e−02
7e−02
1e−01
2e−01
3e−01
5e−01
7e−01


F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate
Selectivity, SEL(T)
7e+01
5e+01
3e+01
2e+01
1e+01
7e+00
5e+00
3e+00
2e+00
1e+00
7e−01
5e−01
3e−01
2e−01
1e−01
7e−02
5e−02
3e−02
2e−02
1e−02
7e−03
5e−03
3e−03
2e−03
1e−03
7e−04
5e−04
3e−04
2e−04
1e−04
7e−05
5e−05
3e−05
2e−05
1e−05
1e−05
3e−05
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
False Positive Identification Rate, FPIR(T)
Enrolled images:
recent
N = 1600000
Mugshot natural
Mugshot webcam

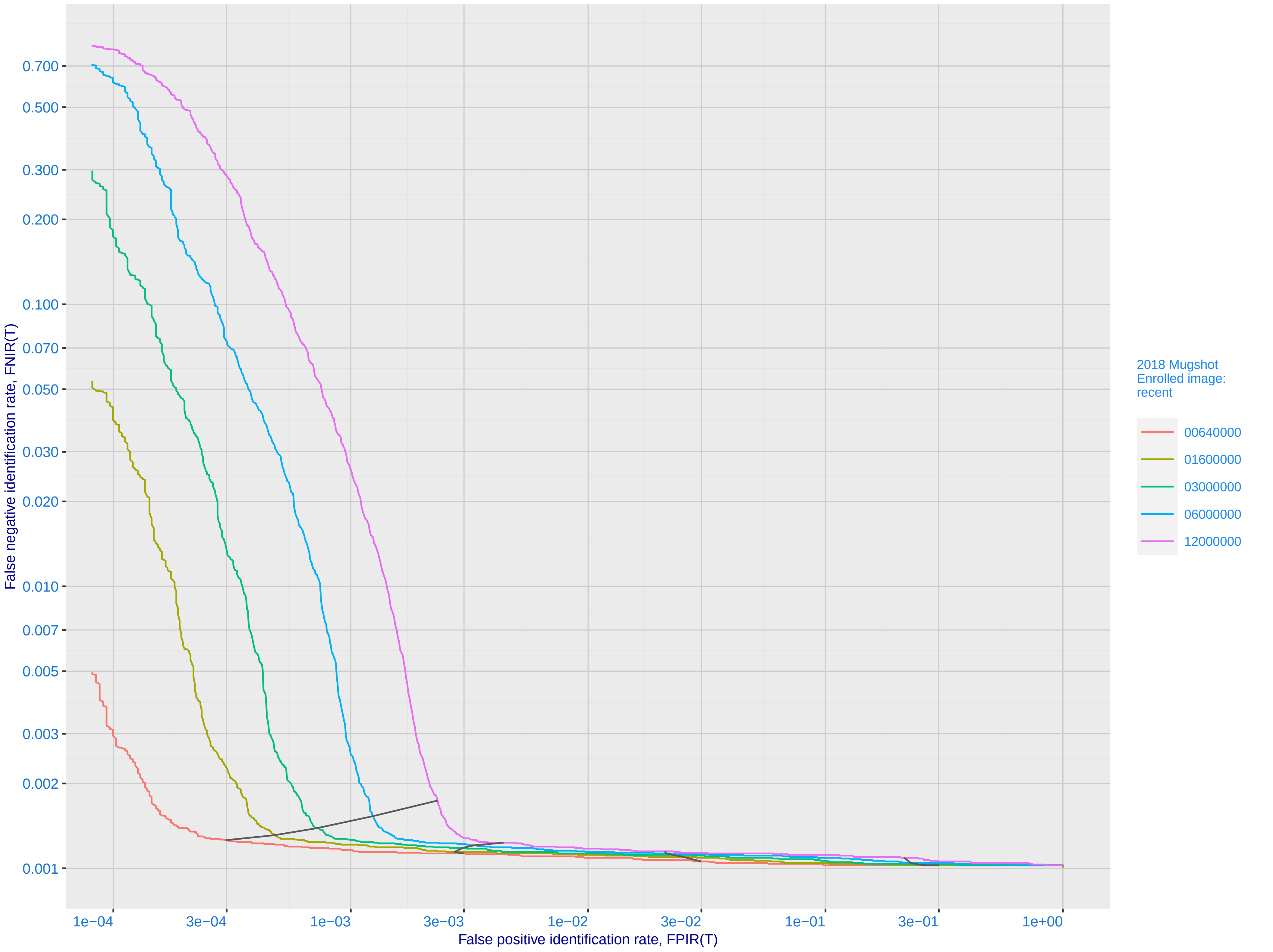

G: FPIR dependence on T by probe type for N = 1600000 subjects
False Positive Identification Rate, FPIR(T)
7e−01
5e−01
3e−01
2e−01
1e−01
7e−02
5e−02
3e−02
2e−02
1e−02
7e−03
5e−03
3e−03
2e−03
1e−03
7e−04
5e−04
3e−04
2e−04
1e−04
7e−05
5e−05
3e−05
2e−05
1e−05
2000
4000
6000
Threshold
Enrolled image:
random
N = 1600000
Immigration visa−border
Immigration visa−kiosk
Mugshot natural
Mugshot profile
Mugshot webcam
enrolment_style
random
recent

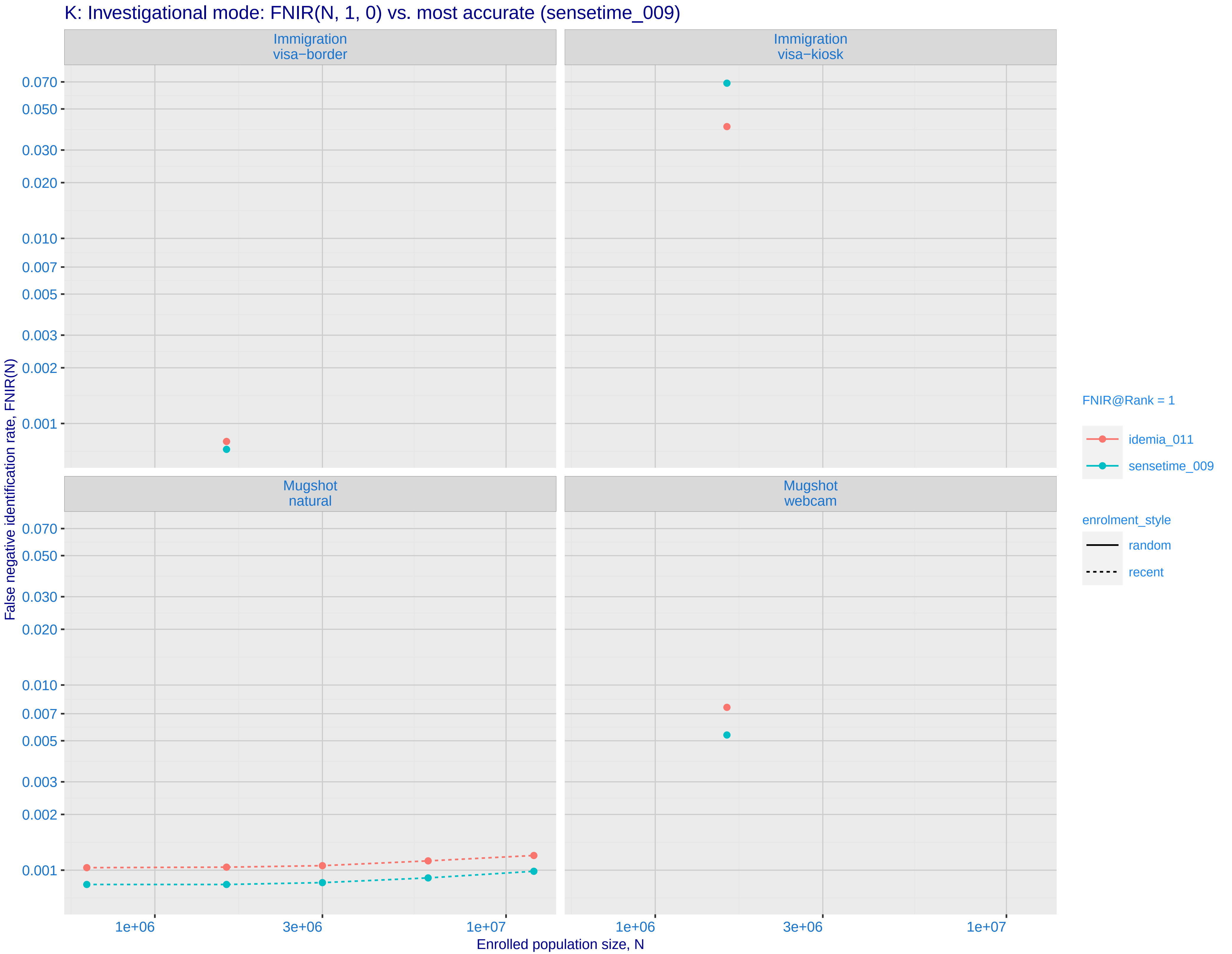

H: Reduced length candidate lists for human review
Dataset is border−border with time−lapse [10,15] YRS with N = 1600000. Probes are 10−15 years later than enrollment image
Mated
without threshold, all lists have same length, here 50
with low threshold, list length shortens
towards 1 as there is 1 mate
at high threshold, lists
gradually get shorter as mates
fall below threshold (misses)
the pink region shows
[5,95]−percentile of list lengths
40
20
10
4
2
1
Non−Mated
without threshold, all lists have same length, here 50
with low threshold, list length shortens
at high threshold, lists only rarely
have any candidates
here only 1 in 100
40.00
20.00
10.00
4.00
2.00
1.00
0.40
0.20
0.10
0.04
0.02
Mean candidate list length
2000
4000
6000
Threshold value



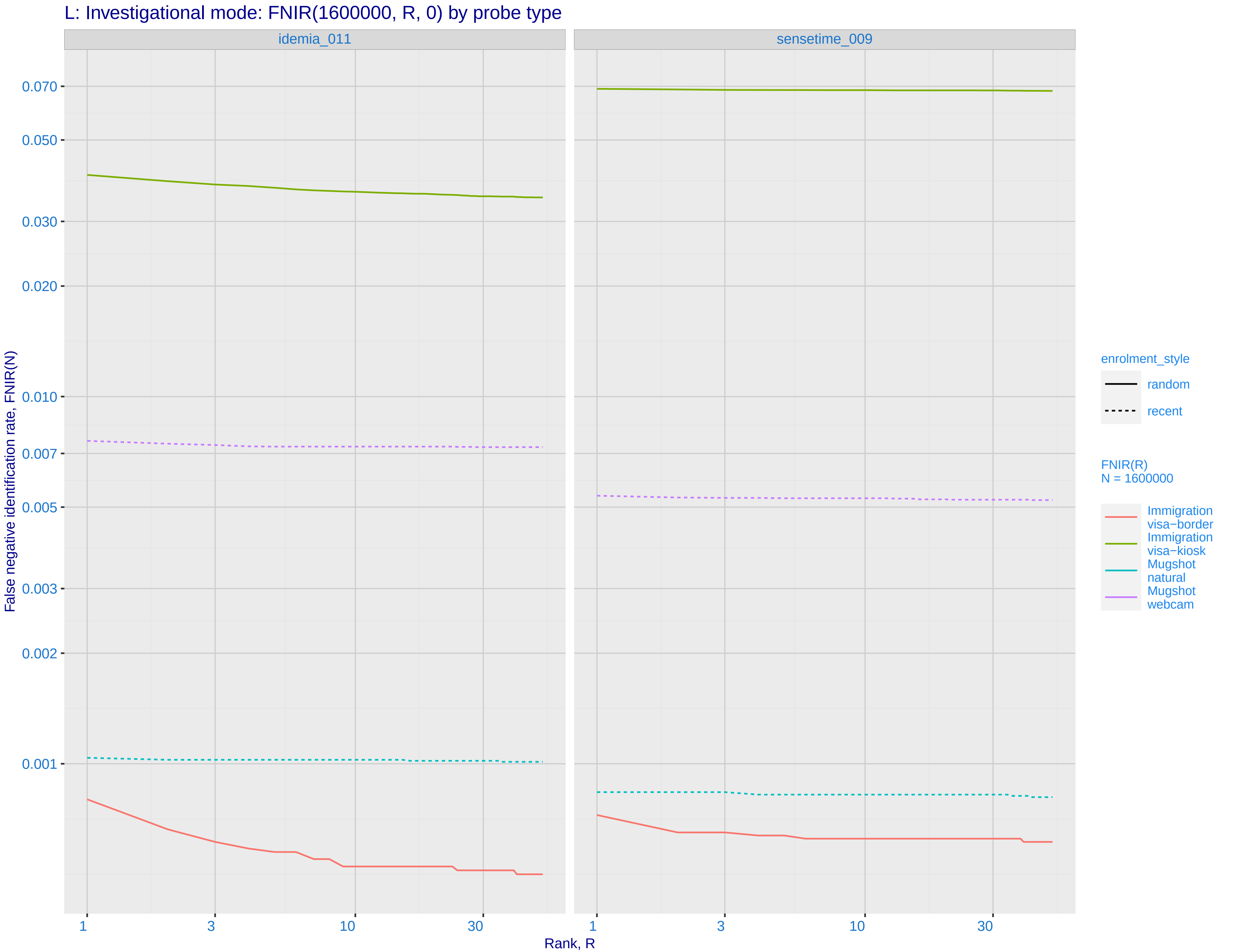
I: FNIR and FPIR dependence on threshold
Dataset is border−border with time−lapse [10,15] YRS with N = 1600000. Probes are 10−15 years later than enrollment image
FNIR(T)
0.400
0.200
0.100
0.040
0.020
0.010
0.004
0.002
FPIR(T)
1e+00
4e−01
2e−01
1e−01
4e−02
2e−02
1e−02
4e−03
2e−03
1e−03
4e−04
2e−04
1e−04
2000
4000
6000
Threshold value
False {Negative | Positive} Identification Rate

J: DET for Mugshot natural images and various N. Links connect points of equal threshold.

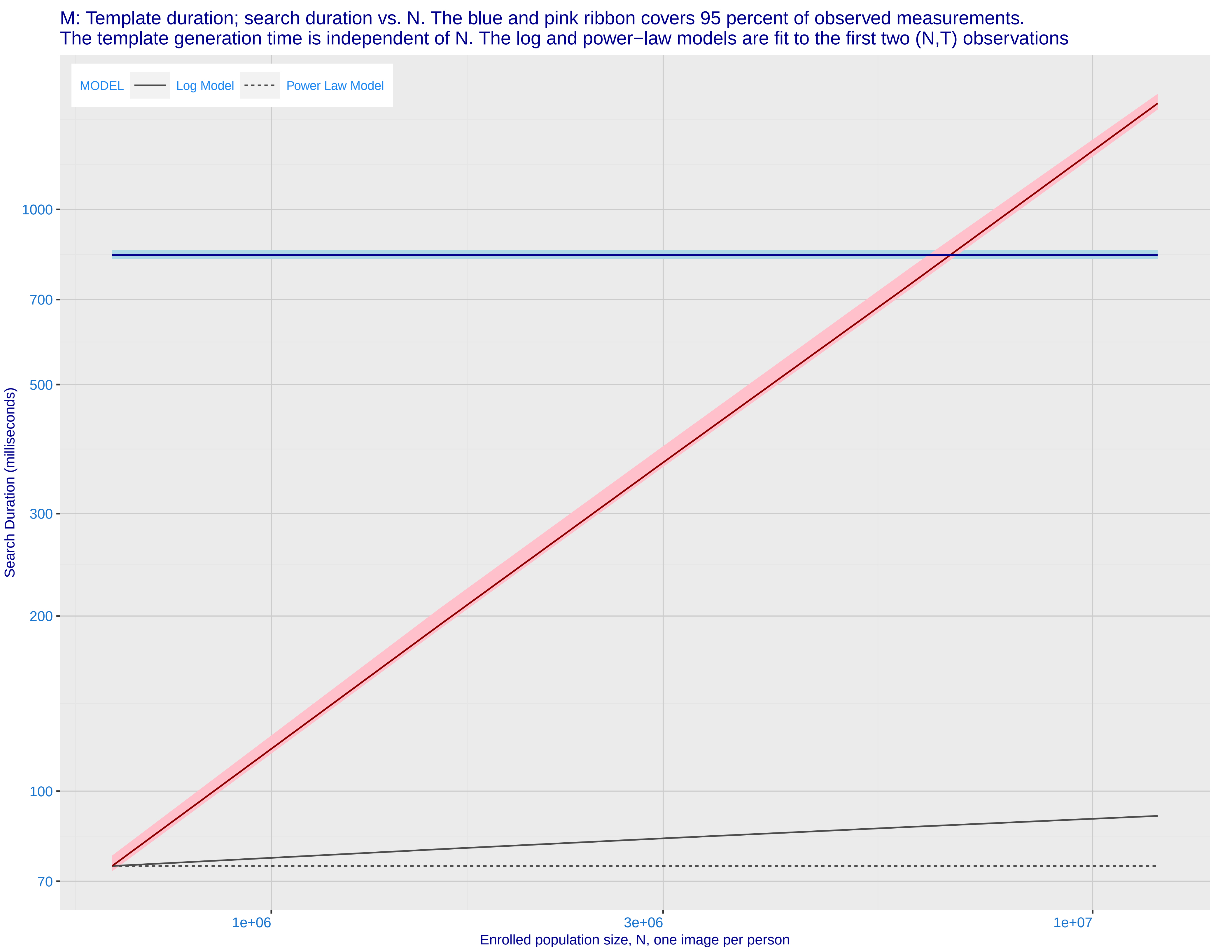




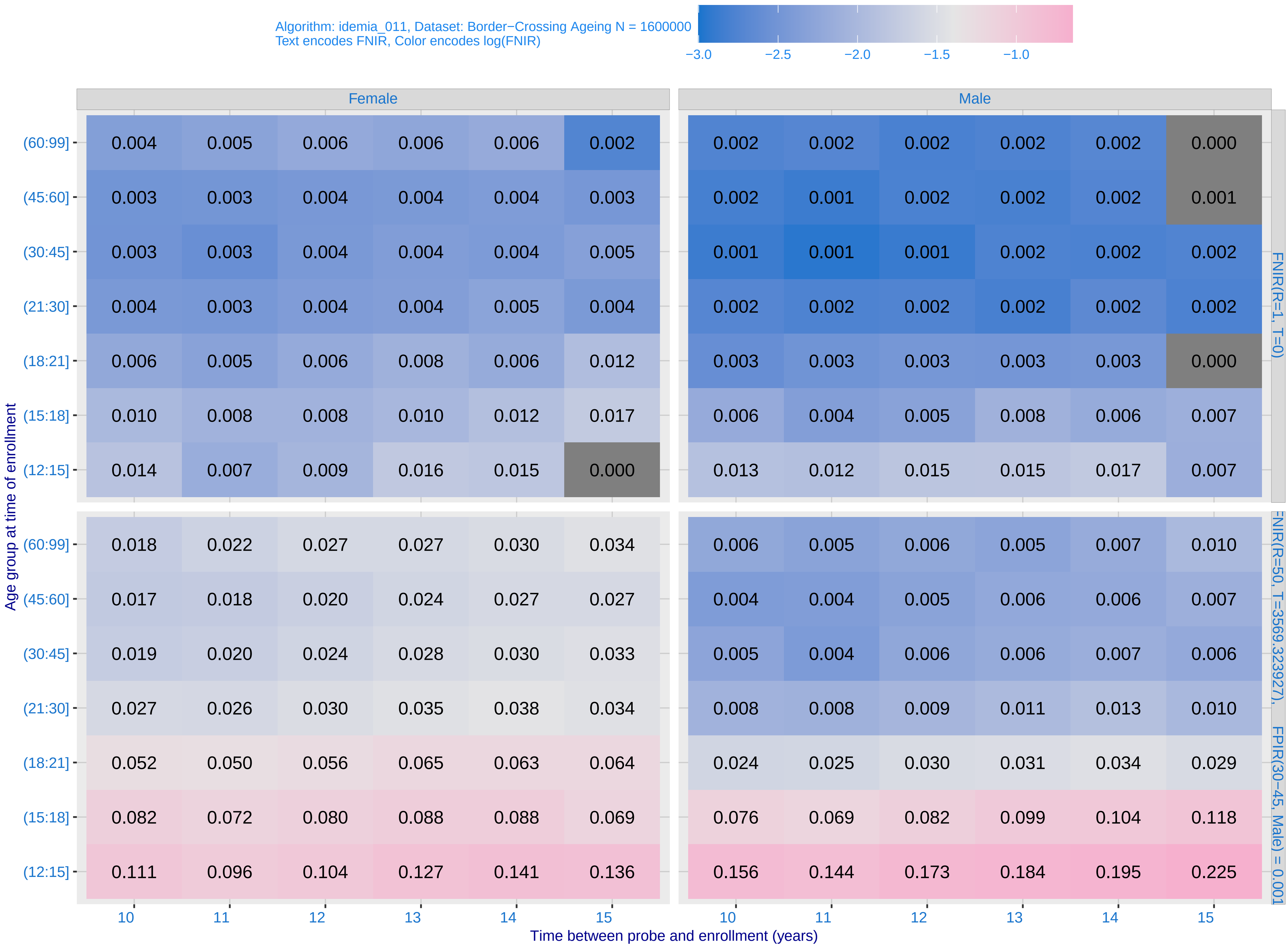
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)
Immigration
visa−border
Immigration
visa−kiosk
Mugshot
natural
Mugshot
webcam
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(N)
1e+06
3e+06
1e+07
Enrolled population size, N
FNIR@Rank = 1
idemia_011
sensetime_009
enrolment_style
random
recent



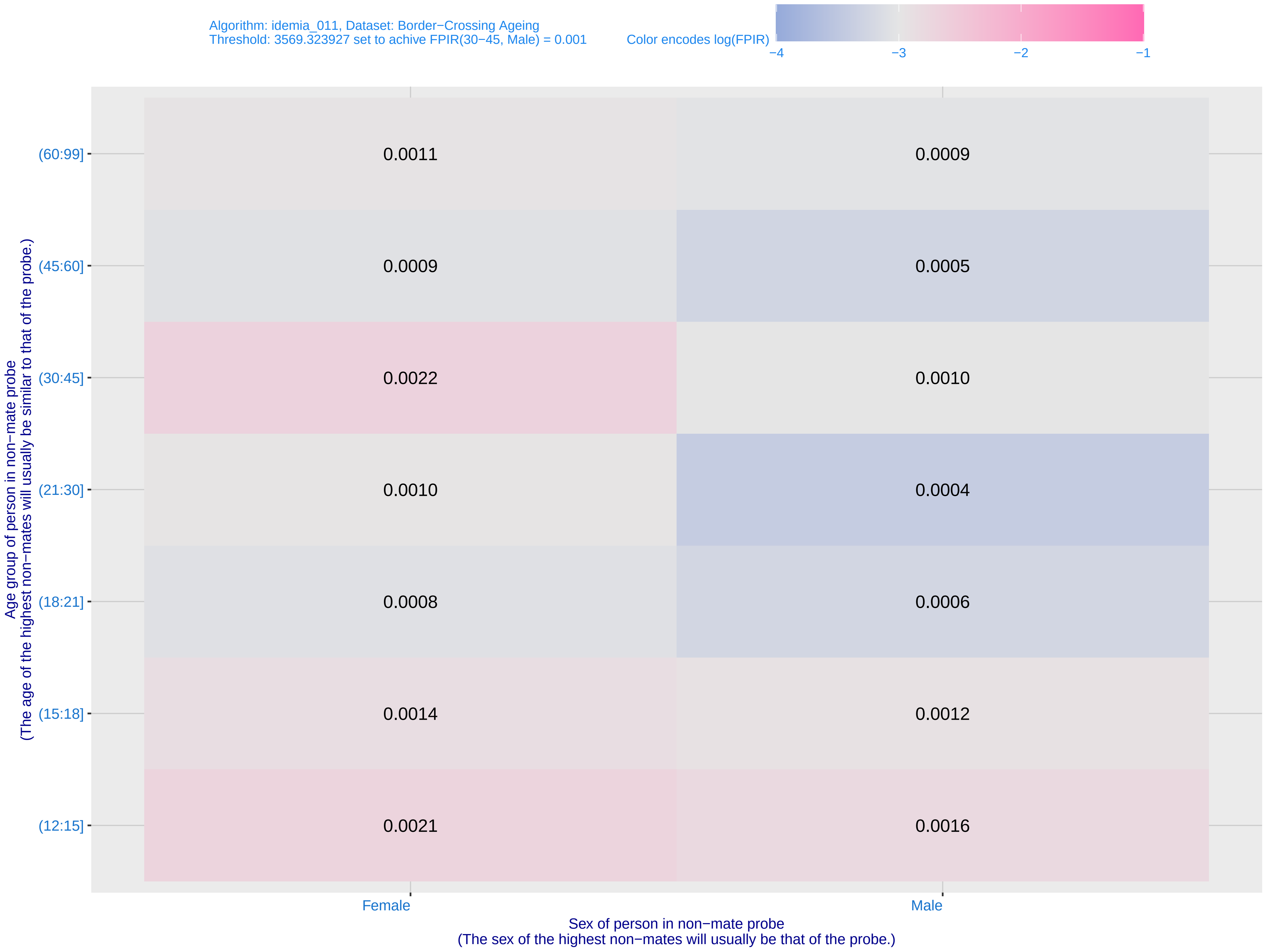
L: Investigational mode: FNIR(1600000, R, 0) by probe type
idemia_011
sensetime_009
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(N)
1
3
10
30
Rank, R
enrolment_style
random
recent
FNIR(R)
N = 1600000
Immigration visa−border
Immigration visa−kiosk
Mugshot natural
Mugshot webcam

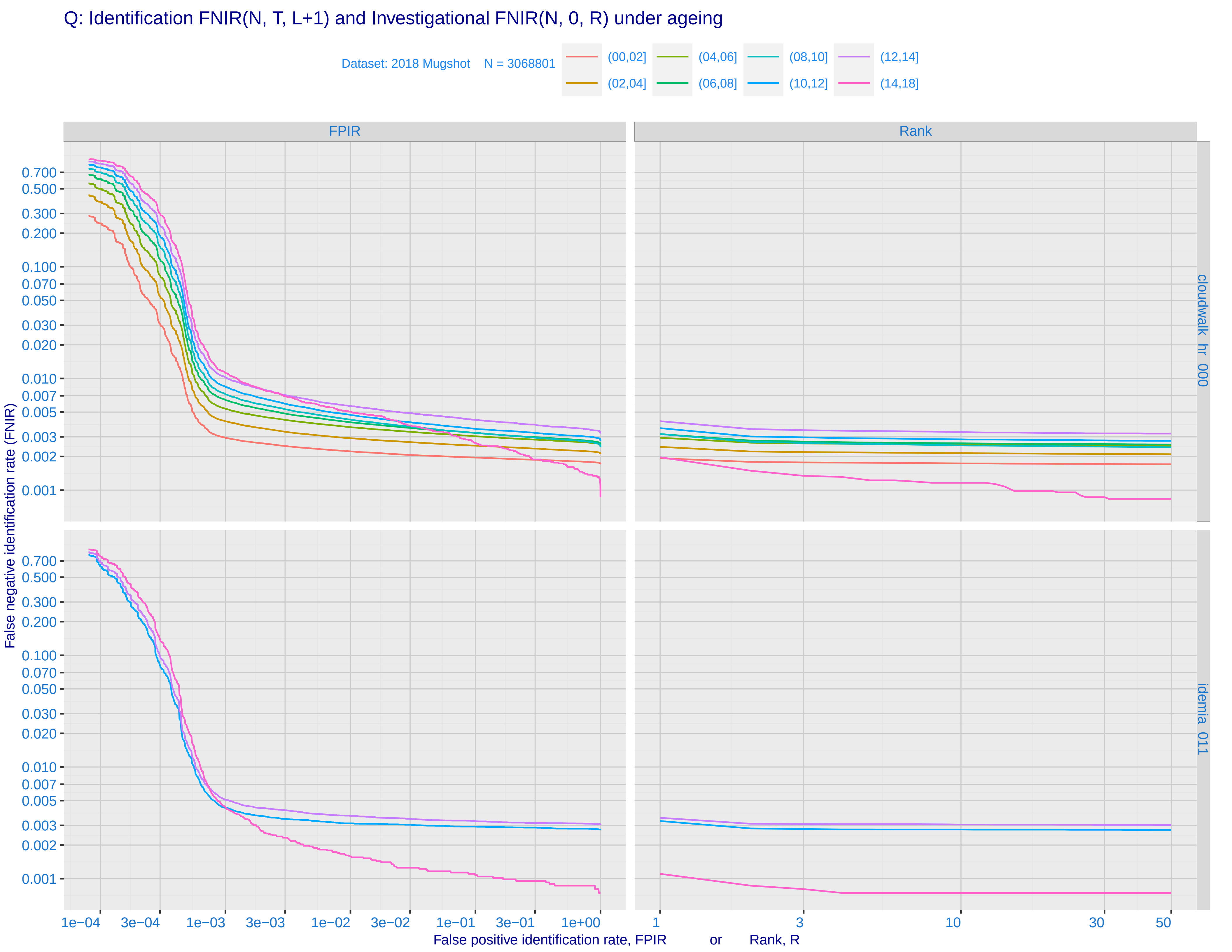

M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements.
The template generation time is independent of N. The log and power−law models are fit to the first two (N,T) observations
MODEL
Log Model
Power Law Model
1000
700
500
300
200
100
70
Search Duration (milliseconds)
1e+06
3e+06
1e+07
Enrolled population size, N, one image per person

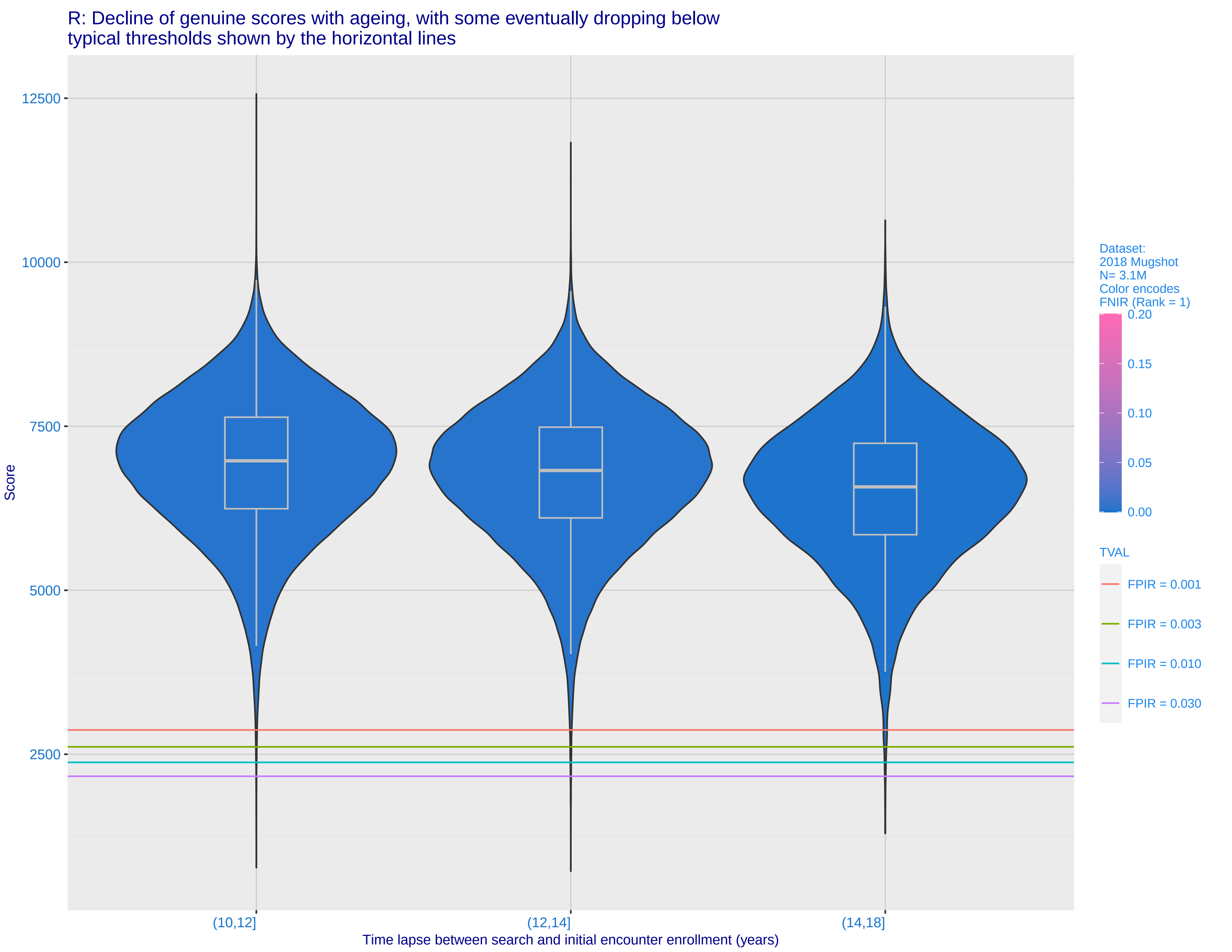
O: FNIR(T, N = 1.6 million) by sex, age and time−lapse. The top row gives investigational rank−1 miss rates.
The bottom panels give high threshold for more lights−out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.




Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing
Dataset: 2018 Mugshot N = 3068801
(00,02]
(02,04]
(04,06]
(06,08]
(08,10]
(10,12]
(12,14]
(14,18]
FPIR
Rank
cloudwalk_hr_000
idemia_011
False negative identification rate (FNIR)
0.700
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
1e+00
1
3
10
30
50
False positive identification rate, FPIR or Rank, R



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines
12500
10000
7500
5000
2500
Score
(10,12]
(12,14]
(14,18]
Time lapse between search and initial encounter enrollment (years)
Dataset:
2018 Mugshot
N= 3.1M
Color encodes
FNIR (Rank = 1)
0.20
0.15
0.10
0.05
0.00
TVAL
FPIR = 0.001
FPIR = 0.003
FPIR = 0.010
FPIR = 0.030